



AI-Predictive Maintenance for Smart City Water Distribution Networks

1 ABSTRACT



Water distribution networks in smart cities face significant challenges due to undetected leaks and reactive maintenance, leading to high water losses and increased operational costs. This project proposes an AI-based predictive maintenance system that leverages machine learning models to detect leaks early and accurately localize their positions accurately. The system enables proactive decision-making, reduces water loss, and supports sustainable and intelligent urban water management.

2 OBJECTIVES



- ✓ Detect leaks early in the water distribution network.
- ✓ Identify the location of leaks accurately.
- ✓ Reduce water loss and maintenance costs.
- ✓ Support smart city infrastructure with intelligent, real-time decision-making.

3 MODELS

MODEL 1 - LEAK DETECTION

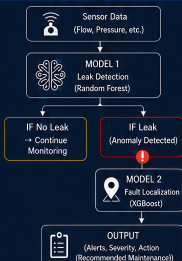


Benchmark Results

Model	Recall	F1-Score	ROC-AUC
Random Forest	0.9876	0.9937	0.9999
XGBoost	0.9876	0.9907	0.9999
SVM	0.9722	0.8122	0.9954
KNN	0.9411	0.7077	0.9773

★ Selected Model: Random Forest

SYSTEM INTEGRATION



3 MODELS

MODEL 2 - FAULT LOCALIZATION



Benchmark Results

Model	Accuracy	F1-Macro
XGBoost	0.9537	0.9541
Random Forest	0.9629	0.9634
Logistic Regression	0.9536	0.9537
Decision Tree	0.9381	0.9382
SVM	0.8173	0.8178
KNN	0.7398	0.7393

★ Selected Model: XGBoost

4 METHODOLOGY

-  Data Preprocessing
Clean, transform, and handle missing data.
-  Feature Engineering
Extract meaningful features from sensor data.
-  SMOTE
Address class imbalance using SMOTE.
-  Model Training
Train models using optimized parameters.
-  Stratified Cross-Validation
Evaluate models using Stratified

5 DASHBOARD & SYSTEM OUTPUT



ALERT SHARING SYSTEM



6 CONCLUSION

- ✓ The proposed system effectively detects leaks with high accuracy using machine learning.
- ✓ Fault localization is reliable and supports targeted maintenance.
- ✓ A multi-model architecture improves overall performance and robustness.
- ✓ Real-time alerts and web alert sharing enable operators response time and operational efficiency.
- ✓ The system contributes to smart city goals by reducing water loss and supporting sustainable water management.

7 TEAM MEMBERS

1. Alanoud Abdullah Alsaqabi
431201832

2. Hailah Ayed Alshammari
431201535

3. Reema Ali Alsaud
431217986

SUPERVISOR
Dr. Murad Rassam